

1. The simplest idea (no math yet)

Imagine price data as a line of people standing in a room.

Case A: stable market

Everyone is clustered around the middle.



The “center” of the group doesn’t move much.

Case B: drifting structure (important idea)

Now imagine something subtle happens:

- people are still close together (no obvious trend)
- BUT the whole group slowly shifts upward



You might NOT see a strong trend yet, but something is happening:

👉 the *center of mass* is moving

This is what your metric is trying to detect.

2. What is “Central Tendency”?

In simple terms:

- **Mean (μ)** = average person location
- **Median (m)** = middle person location

Both are ways of answering:

“Where is the center of this group?”

3. What is “distortion” here?

Now imagine:

- the mean and median should normally be close

- if they start separating → something changed in shape

So:

Normal situation

mean $\approx$ median

Distorted situation

mean $\neq$ median

That difference is the first signal.

4. The key formula (intuitive version)

You defined:

$$C = \frac{|\mu - m|}{S}$$

Let's translate:

Numerator: $|\mu - m|$

👉 "How far apart are the two definitions of center?"

Think:

"Is the center consistent or being pulled around internally?"

Denominator: S (scale)

This is very important.

S = average step movement

Think of it as:

"How much does price normally wiggle per step?"

So it gives context.

5. Why divide by S?

Without scaling:

- a big market = naturally bigger differences
- a quiet market = naturally smaller differences

So we normalize.

Final meaning:

“How big is the center distortion compared to normal movement size?”

6. Intuition of the whole metric (VERY IMPORTANT)

C answers this:

“Is the market’s *center* stable relative to how much it usually moves?”

7. Simple analogy (best one)

Imagine a classroom of students.

Stable system:

Students are sitting in chairs.

- average position = center of room
- median position = same
- no one is shifting the structure

👉 $C \approx 0$

Distorted system:

Now imagine:

- some students slowly shift to the front rows
- others stay behind
- no obvious “trend direction”, just internal reshuffling

Now:

- mean moves forward
- median lags behind

👉 C increases

But importantly:

⚠️ nobody is running or screaming (no volatility spike)

⚠️ it is *structural change*, not noise

8. Key conceptual separation (this is critical)

You already had another metric (symmetry / imbalance).

So now you have TWO different ideas:

Concept	What it detects
Symmetry	who is winning (buyers vs sellers)
C (this metric)	how the <i>shape of the distribution</i> is changing internally

9. What this is REALLY trying to detect

This is the most important part:

C is NOT about direction

It is about:

“Is the market quietly reshaping itself before it moves?”

10. One-line summary

If we compress everything:

C measures how much the “center of the market” is drifting relative to normal movement size.

11. Why this matters (intuition in trading terms)

Markets often do this:

1. first: structure slowly shifts (C rises)
2. later: imbalance appears (symmetry changes)
3. then: trend becomes visible

So C is:

- 👉 early structural warning
- 👉 before trend is obvious